Long-Context Language Models Require Extreme Sparsity in Context Dimension

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Abstract

Sparsity has long been a central theme in LLM efficiency but its role in context processing remains unresolved. As LLM workloads shift toward longer contexts and agentic interactions, the compute and memory bottlenecks of attention become increasingly critical, raising the question of whether these constraints are fundamental. Our position: this constraint is artificial, unnecessary and the future of LLM inference lies in extreme but principled sparsity along the context dimension. Our position comes out of various bits of empirical and theoretical evidence. Firstly, we find insistance on dense attention unreasonable since in long context a query essentially projects the O(N) attention information into hidden space of diemsion $d << N$. This, as we show is extremely lossy. Instead, we propose a completely context-sparse context processing combining top-k sparsity with linear attention / SSMs as the way forward. To support our proposal we show that this combination can approximately cover the functional space of dense softmax attention. Independently but importantly, in a first elaborate study of its kind, we empirically show a strong trend towards current LLM models, which are not trained for context sparsity, being extremely robust to inference time decode-sparsity across tasks of varying complexities such as retrieval, multi-hop QA, and mathematical reasoning and agentic coding. For instance, Qwen3.5-27B can tolerate upto 100x sparsity on benchmarks of RULER-HARD, LOFT and AIME2025 without loss of quality, and upto $50\times$ on SWE with a small drop in quality. These results emphasize the possibility that we can transition to complete sparsity without any loss of capability. Additionally, we also discuss what this shift in paradigm of context processing means for hardware. Importantly, we show that even current hardware is equipped enough to realise gains from this sparsity. For instance, our sparse decode kernels can accelerate large context processing by a factor 10x over FlashInfer at 50x sparsity levels on current hardware such as H100. Overall, these results position extreme context sparsity not as a heuristic, but as a principled foundation for LLM inference, training, and architecture design, both feasible and beneficial, and a compelling direction for future systems.

1 Introduction

Sparsity in LLM inference has long been a goal for researchers. Efforts to sparsify the FFN component of Transformers [5, 12, 33, 40] have largely converged on Mixture-of-Experts (MoE) as the de facto architecture across frontier labs [2, 29, 41, 47, 55]. In contrast, a similar consensus has yet to emerge for attention mechanisms, or more broadly, context processors. Research on sparsity in attention dates back several years and has recently re-emerged in the LLM era, primarily focusing on emergent sparsity in trained models [6, 9, 10, 11, 16, 22, 26, 32, 43, 49, 56, 57, 58]. However, such sparsity is rarely adopted in state-of-the-art inference engines, highlighting a limited understanding of its

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practical value. There has been some consolidation of this line of work through the adoption of sparse attention during post-training [30, 55]. However, demonstrated gains are largely confined to extremely large model regimes, raising questions about their general applicability. In parallel, an alternative form of sparsity has emerged through the development of lightweight context processors, such as linear attention [7, 23, 24, 35, 54], and SSMs [14, 25, 51, 53]. While these approaches have seen some adoption, modern architectures still retain full scaled dot-product attention (SDPA) layers, underscoring the limited expressivity of purely SSM-based models. This leads to a fundamental question: are the quadratic compute bottleneck during prefill and the linear memory bottleneck during decoding inherent constraints that are here to stay?

This question is becoming increasingly important as LLM workloads shift toward substantially longer contexts and generations. Emerging use cases such as agentic tool use, code generation, retrieval augmented generation, multi document reasoning, long form dialogue, and repository scale software understanding are all driving this trend [15, 17, 19, 20, 28, 36, 37, 38, 39, 46, 50]. To ground this in concrete terms, a single 50 page PDF can contain on the order of 33K tokens. An LLM that aims to condition its responses on corporate or legal documents may therefore need to process hundreds of thousands, if not millions, of tokens within a single context window. At the same time, expectations for state of the art models continue to increase, with each generation pushing toward larger and more persistent context handling. Recent examples further highlight this shift. Systems such as openclaw [42] have been reported to use system prompts reaching 160K tokens [1] , illustrating how rapidly context sizes are expanding in practice. Importantly, these workloads do not merely involve longer prompts, but also often require long form generation over extended histories. This challenge is further amplified in agentic settings, where models iteratively generate outputs, incorporate new information, and repeatedly condition on growing interaction traces among agents. As a result, the question becomes of utmost importance: what does the future of LLM Inference looks like in context dimension?

We take a strong position: ***The future of LLM inference lies in extreme sparsity along the context dimension, but this sparsity cannot be naive.*** Our position is supported by three complementary lines of evidence.

Sparsity already exists in new generation models: In section 4 we observe empirically that sparsity is already emerging in modern architectures. Larger and more recent models exhibit remarkable robustness to aggressive context sparsification across a wide spectrum of tasks as shown in Section 4. This holds for relatively simpler benchmarks such as RULER-HARD, a challenging subset of RULER [18], and LOFT [27], as well as for more complex reasoning tasks such as AIME [8] and real-world agentic workloads such as SWE [21] (50+ agentic turns). To our knowledge, this is the first study to examine inference-time sparsity on an agentic workload. Of course, inducing sparsity during training is ideal, but the emergence of sparsity at inference time reinforces the possibility that models can operate effectively in highly sparse regimes and provides greater confidence in designing training procedures that explicitly target sparsity.

We empirically evaluate five model families, including Llama3[13], Qwen2.5[48], Qwen3.5[45], Gemma3[44], and Ministral3[31], across four tasks: RULER-HARD (32K), LOFT (32K and 128K), AIME (65K generation length), and SWE (50+ agentic turns). Our results show that the effectiveness of inference-time sparsity improves with both model scale and the use of hybrid architectures. In particular, larger models from newer-generation families such as Qwen3.5, Gemma3, and Ministral3 sustain quality parity with dense execution even at 50× sparsity. For smaller standard models, the observed degradation can largely be mitigated through stochastic index selection [9]. Notably, these gains are achieved purely at inference time, suggesting that incorporating sparsity during training could yield even greater benefits.

Non-existence of a truly dense attention and a proposal for extreme sparsity: We first show that truly dense attention does not exist in practice: full attention is ultimately bottlenecked by the hidden dimension, causing it to collapse unable to distinguish between varying attention distributions. While the result itself is straightforward, the implications are significant. It suggests that complete sparsity is not merely a practical approximation but principally a superior objective.

[1] https://github.com/openclaw/openclaw/issues/21999

Further, we propose a theoretically grounded framework for complete context sparsity (i.e., a proposal free of any dense SDPA layers). The proposal is provably expressive enough to capture diverse attention regimes of interest and, in principle, enables model architectures that avoid the fundamental bottlenecks associated with standard SDPA. There are two primary schools of thought and corresponding lines of work on "sparsity" in context processing: sparse attention in all layers (DeepSeek [30], GLM [55]) and replacing attention with state-space models [14], including linear attention [24]. We identify that both approaches have negative inexpressibility results. By categorizing functions represented by attention over different key and query distributions into low-entropy and high-entropy functions, we find that sparse attention cannot represent relatively high-entropy functions, while linear attention cannot represent low-entropy functions. Thus, neither approach can solve the problem in isolation. To address this, we propose combining sparse attention with linear attention or SSMs as the appropriate architectural mix. To this end, we show that sparse attention approximates low-entropy functions, while linear attention accurately approximates high-entropy functions, thereby approximately covering the range of entropy achievable by standard full SDPA.

Sparsity can be leveraged for system gains: The alignment of sparsity with hardware is of paramount importance for fully realizing its benefits. It is therefore essential to evaluate whether sparsity actually alleviates the underlying bottlenecks. A common belief is that efficiency gains require block-structured sparsity [59]; without it, sparsity is unlikely to translate into real speedups. We present an argument against this view.

Notably, DeepSeek Attention [30] demonstrates both training- and inference-time speedups with token-level sparsity during prefill. This serves as evidence that token-level sparsity can indeed translate into practical efficiency gains without imposing excessive structure on the model. We extend this further to decoding. For decoding, we provide kernels that accelerate inference under per-token, per-query, and per-head sparsity, i.e., highly irregular sparsity patterns. These gains persist even under grouped query attention, where the number of query heads can be a factor (typically 4) larger than the number of key-value heads. This is possible because the KV cache's vector dimension provides sufficient contiguous memory to make such sparsity effective on modern hardware. In particular, our optimized sparse attention kernel, built on top of FlashInfer with a paged KV-cache backend, achieves up to $10\times$ kernel speedup at $50\times$ sparsity for large batch sizes.

Overall, this paper argues that the community should actively explore extreme sparsity along the context dimension without compromises such as partially retaining full attention layers. The empirical emergence of sparsity across multiple axes highlights strong potential, theoretical insights in Section 3 outline a viable path forward, and decode-time sparse kernel analyses in Section 4, augmented by results from recent DeepSeek models, demonstrate the significant efficiency gains that can be unlocked.

2 Alternative Views

Recent model design trends suggest following alternative views:

View 1: Dense attention layers cannot be excluded. A prevailing school of thought holds that SDPA is necessary for state-of-the-art capability; despite many architectural shifts, SDPA persists even in modern hybrid models. We argue otherwise on two grounds: (i) trained SDPA in state-of-the-art models already exhibits emergent sparsity at inference time across tasks of varying complexity (Section 4); and (ii) SDPA over extremely long contexts cannot remain dense in any meaningful sense, since the attention output is bottlenecked by the hidden dimension and collapses as context grows (Thm. 1).

View 2: Top-k sparsity is all you need. A different line incorporates sparsity uniformly across all attention layers [60], exemplified by DeepSeek-V3.2's lightweight token selection [30]. This paradigm has so far been demonstrated only at extremely large model scales, raising the concern that scale's extra depth and heads are masking the expressivity lost to sparsity rather than complementing it. More fundamentally, deterministic top-k is expressively limited to retrieval and cannot represent higher-entropy distributions that SDPA captures (Cor. 7).

View 3: SSMs and linear attention are all you need. A third school replaces SDPA entirely with linear attention or state-space models [14]. These architectures bring real efficiency gains, but two facts argue against wholesale replacement: (i) frontier-lab models that include these components consistently retain a small number of full attention layers [1, 45], suggesting attention captures something irreplaceable; and (ii) compressed-hidden-state computations are provably incapable of representing the low-entropy retrieval-like distributions SDPA expresses [4] (Section 3).

3 Proposal for Complete Context Sparsity

We begin by examining whether attention layers can remain truly dense as context length grows, and show that they cannot. We then explore how sparse attention and linear attention/SSM are complementary by nature, and propose their combination as a principled path forward.

3.1 Dense Attention Collapses Through The Hidden Dimension.

While sparse attention and SSMs have their own shortcomings, fully dense attention has its own bottleneck: it assigns a weight to every context token, but the layer passes forward only a fixed-dimensional hidden vector.

Theorem 1. *Let $V \in \mathbb{R}^{N\times d}$ be any value matrix, and let the dense attention output be $o = V^\top a$, where $a = (a_1, \dots a_N)$ is an attention distribution over N context tokens. If $d < N - 1$, then this map is not injective on the attention simplex. In particular, there exist two distinct dense attention distributions a, a' such that $a \neq a'$, and $V^\top a = V^\top a'$. Thus, a d-dimensional post-attention embedding cannot preserve all dense attention-score variations over more than $d + 1$ context tokens.*

Corollary 2. *If all dense attention distributions over N context tokens must remain distinguishable after the attention output $V^\top a$, then the hidden dimension must satisfy $d \geq N - 1$. Thus, losslessly preserving arbitrary dense attention over a million-token context requires million-scale hidden width.*

This is the embedding bottleneck. When $d \ll N$, the map $a \mapsto V^\top a$ collapses many distinct attention patterns into the same hidden representation; hence not every fine-grained variation in a dense attention distribution can be carried forward by the post-attention vector. This conclusion is consistent with lower bounds for indexed lookup: finite-precision recurrent models, including RNNs, LSTMs, state-space models, and recurrent linear attention, require hidden-state size $\Omega(N)$ to recover arbitrary tokens from an N-token sequence [4]. Together, these observations motivate a context-sparse decomposition: sparse attention for query-dependent retrieval, and compact mechanisms for diffuse aggregation.

3.2 A Unifying View of Sparse Approximation and Linear Attention

We now formalize this context-sparse decomposition. The key distinction is whether a query needs to retrieve a small number of specific tokens or aggregate information spread across many tokens. Sparse SDPA is naturally suited to the first case, while linear or state-based mechanisms are better suited to the second. We begin by fixing notation for dense attention and its sparse restriction.

Dense Attention. For a query $q \in \mathbb{R}^d$, keys $k_1, \dots, k_N \in \mathbb{R}^d$, and values $V \in \mathbb{R}^{N\times d}$, we write the dense SDPA weights and output as $a_i := \frac{\exp(q^\top k_i/\sqrt{d})}{\sum_{j=1}^N \exp(q^\top k_j/\sqrt{d})}$ and $o := V^\top a = \sum_{i=1}^N a_i v_i$.

Sparse Attention. Let $S \subset [N]$ denote the selected subset of context tokens. We define the sparse attention weights and the output by $\widetilde{a}_i := \frac{a_i}{\sum_{j\in S} a_j} \mathbf{1}\{i \in S\}$ and $\widetilde{o} := V^\top \widetilde{a} = \sum_{i\in S} \widetilde{a}_i v_i$.

Definition 3. *For a query, let $a = (a_1, \dots, a_N)$ be its softmax attention distribution over N context tokens. For a subset $R \subset [N]$, define the retrieval mass of a on R as $\rho_R(a) := \sum_{i\in R} a_i$. We say that a performs (m, ρ)-retrieval if there exists a subset $R \subset [N]$ with $|R| = m$ such that $\rho_R(a) \geq \rho$. Equivalently, if $M_m(a)$ denotes the mass of the largest m entries of a, then a performs (m, ρ)-retrieval if and only if $M_m(a) \geq \rho$.*

Definition 4. *Fix $m \in [N]$ and $\rho \in [0, 1]$. We say that an attention distribution a is (m, ρ)-diffuse if no subset of m tokens captures mass ρ, i.e., $\max_{|R|=m} \sum_{i\in R} a_i < \rho$. Equivalently, a is (m, ρ)-diffuse if $M_m(a) < \rho$.*

3.2.1 Retrieval Mass Controls Sparse Approximation

Definition 3 captures sharp token-level lookup: a small set of context positions contains most of the information needed by the query. Definition 4 captures broad aggregation: no small set of tokens dominates, so deterministic top-k truncation is not the right abstraction. The next result connects this distinction to output error by showing that sparse attention is accurate when the selected set captures most of the dense attention mass.

Theorem 5. *For any subset $S \subset [N]$, let $\delta(S) := \sum_{i \notin S} a_i$ denote the missed tail-mass. and $\widetilde{a}$ be the corresponding sparse attention distribution over S . Then $\|o - \widetilde{o}\|_2 \leq \sqrt{2}\,\|V\|_2\,\delta(S)$. In particular, if S is the set of top-k entries of a, then $\|o - \widetilde{o}\|_2 \leq \sqrt{2}\,\|V\|_2\,(1 - M_k(a))$.*

The following two results are direct consequences of Theorem 5.

Corollary 6. *For (k, ρ)-retrieval, top-k sparse attention satisfies $\|o - \widetilde{o}\|_2 \leq \sqrt{2}\,\|V\|_2\,(1 - \rho)$.*

Corollary 7. *If a is (k, ρ)-diffuse, then every deterministic k-token sparse set misses more than $1 - \rho$ attention mass. In particular, no deterministic k-token truncation can be certified by Theorem 5 to have error smaller than $\sqrt{2}\,\|V\|_2\,(1 - \rho)$ using mass capture alone.*

Note that Corollary 7 does not say that sparse attention is futile in diffuse regimes. Rather, it says that deterministic small-budget top-k is the wrong abstraction when useful information is distributed across many tokens.

3.2.2 Linear Attention as an Aggregation Mechanism

Section 3.2.1 shows that sparse attention is appropriate for retrieval-like distributions and that deterministic small-budget truncation is not certified when the distribution is diffuse. We now analyze the complementary regime, where no small set of keys is sharply separated from the rest. To make this aggregation side concrete, we analyze positive FAVOR+[7] as a representative random-feature linear attention mechanism as it provides a clean theoretical model that approximates the softmax kernel with existing second-moment guarantees. This lets us formally show that, under bounded query–key alignment, a linear random-feature mechanism can approximate the dense softmax output in the diffuse regime.

Theorem 8. *For $0 \leq \tau \leq 1$, assume that $\|q\|_2 = \|k_i\|_2 = 1$ and $|q^\top k_i| \leq \tau$ for $i = 1, \ldots, N$. Let positive FAVOR+ be applied with m independent random features to $x = q/d^{1/4}$, $y_i = k_i/d^{1/4}$, so that $\widehat{K}_i = \phi(x)^\top \phi(y_i)$ is an unbiased estimator of $K_i = \exp\left(q^\top k_i/\sqrt{d}\right)$. Define the random-feature attention weights and output by $\widehat{a}_i := \widehat{K}_i / \sum_{j=1}^N \widehat{K}_j$ and $\widehat{o} := V^\top \widehat{a}$. Then, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\|o - \widehat{o}\|_2 \leq 2\sqrt{\frac{\exp\left(2 + 2\tau/\sqrt{d}\right) - 1}{m\delta}}\,\|V\|_2.$$

Connection to Attention Entropy: The bounded-alignment condition also clarifies why Theorem 8 corresponds to an aggregation-like regime. For a probability distribution $p = (p_1, \ldots, p_N)$, let $H(p) := -\sum_{i=1}^N p_i \log p_i$ denote its entropy. Note that the assumption in Theorem 8 directly implies $\max_i q^\top k_i/\sqrt{d} - \min_i q^\top k_i/\sqrt{d} \leq 2\tau/\sqrt{d}$. Therefore, the softmax weights satisfy $e^{-2\tau/\sqrt{d}}/N \leq a_i \leq e^{2\tau/\sqrt{d}}/N$, for all i. Consequently, $H(a) \geq \log N - \frac{2\tau}{\sqrt{d}}$. Thus, bounded query–key alignment implies that the softmax distribution is diffuse in the sense of Definition 4. The entropy bound is only a diagnostic consequence; the main point is that no key is sharply separated from the rest. A bounded-norm extension ($\|q\|_2, \|k_i\|_2 \leq R$, with factor $\exp((2R^2 + 2\tau)/\sqrt{d})$ in place of $\exp((2 + 2\tau)/\sqrt{d})$) is given in Appendix D.

The proposal in one picture. Theorems 1, 5, and 8 motivate a decomposition of dense context processing. When $N \gg d$, dense SDPA computes attention-distribution variations that the d-dimensional post-attention vector cannot all preserve. Retrieval-like distributions, where a small set of tokens captures most of the mass, are accurately handled by sparse top-k attention with error controlled by missed tail mass. Diffuse aggregation regimes, where no small subset dominates, are naturally matched to random-feature linear attention, with error controlled by query–key alignment. Thus, sparse SDPA and compact aggregation mechanisms cover complementary regimes: deterministic top-k has no mass-capture guarantee in diffuse regimes, while compact recurrent state cannot support arbitrary token-level retrieval without linear scaling [4]. The remainder of the paper provides empirical evidence (Section 4) and systems-level evidence (Section 5.2) that this context-sparse design is feasible and beneficial in practice.

4 Emergent Sparsity Observed in State-of-the-art Models

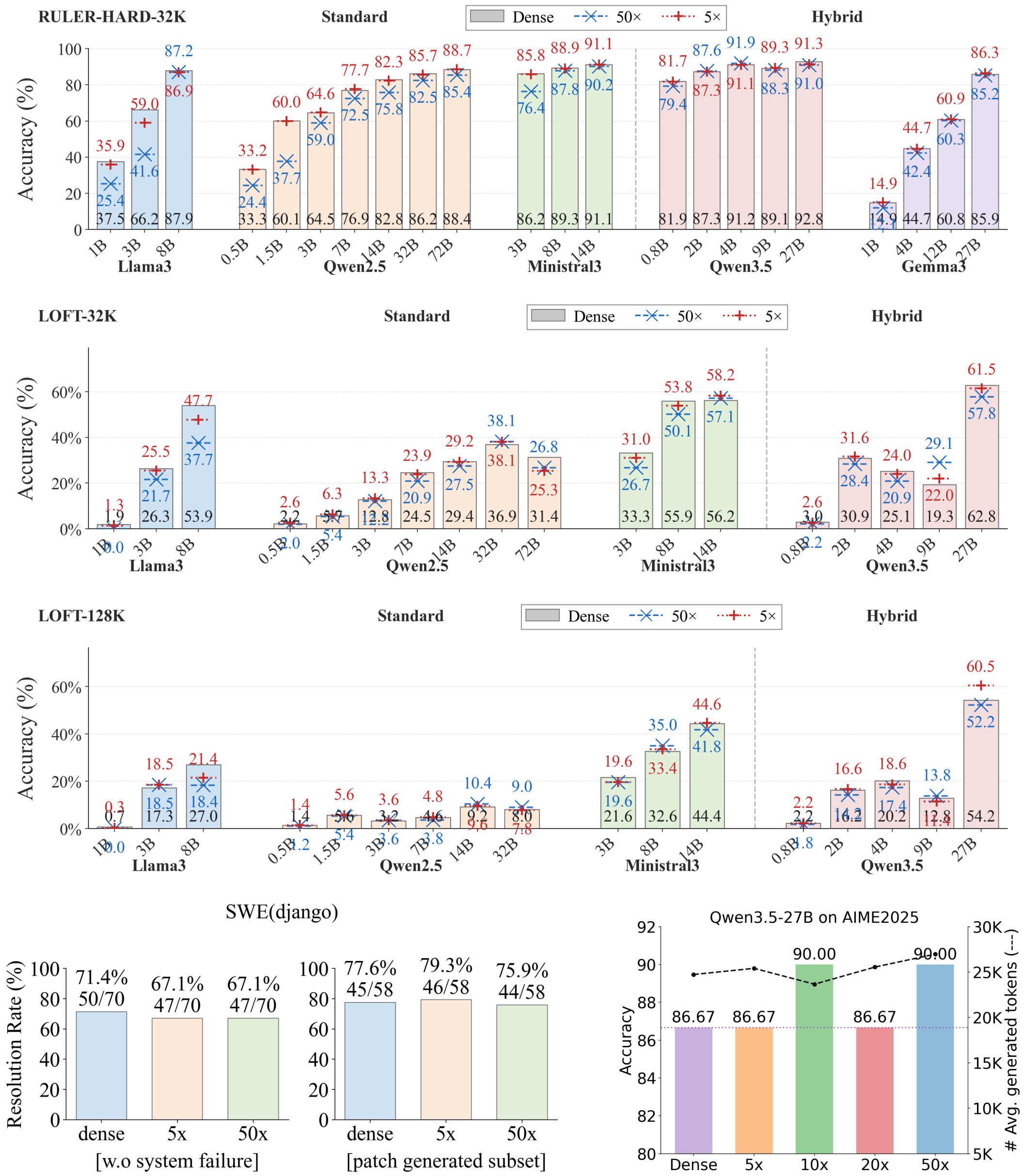

Figure 1: **Inference-time top-k sparsity preserves quality across models and tasks.** Score vs. sparsity ratio K ($1\times$ = dense; $K\times$ = attends to $1/K$ tokens) under oracle top-k selection. Across five model families and five tasks (RULER, LOFT, SWE-Bench Django, AIME-2025), large hybrids (Qwen3.5-27B, Gemma3-27B) remain within ~ 1 point of dense up to $50\times$, while smaller models degrade smoothly. SWE-Bench Django head-to-head: on the 70-task subset where all three configurations attempted (no step-limit hits), dense reaches 71.4% vs 67.1% at $5\times$ and $50\times$; on the stricter 58-task subset where all three submitted a patch, sparse is within ~ 2 points of dense (77.6% dense, 79.3% at $5\times$, 75.9% at $50\times$). Trends are consistent across 32K and 128K; includes the first inference-time sparsity result on an agentic benchmark.

We now turn to empirical evidence showing that even under current training recipes and latest architectures, where attention is not explicitly trained to be sparse, sparsity emerges regardless. This reinforces our proposal: it suggests that in principle, models can transition to extremely sparse context processing without any loss in capability. We adopt the following evaluation setup.

Algorithms: We enforce sparse attention at decode time and measure the effect across models, architectures, and tasks. The primary mechanism is exact *oracle* top-k selection (to eliminate confounds from approximate top-k indexers); in selected experiments we also report stochastic indexing via vAttention [9].

Datasets: For long-context retrieval, we report the average score on RULER-32K-HARD, a challenging subset consisting of six tasks from RULER-32K [18] (`fwe`, `qa1`, `qa2`, `vt`, `nm2`, and `nm3`). On the LOFT [27] benchmark, we evaluate long-context performance by averaging results over five datasets (`hotpotqa`, `nq`, `musique`, `qampari`, and `quest`) at both 32K and 128K context lengths. To assess sparsity in long-form generation settings, we evaluate the largest hybrid model, Qwen3.5-27B [45], on AIME2025 [8], where we limit the generation length to 65K tokens. For agentic workloads, we evaluate Qwen3.5-27B on the SWE-Bench Django [21] subset (114 tasks, ≤ 100 turns per task) at three sparsity levels: dense, oracle-top-20 ($5\times$ sparsity), and oracle-top-2 ($50\times$ sparsity). To our knowledge this is the first inference-time sparsity study on an agentic benchmark.

Models. We investigate five families, with a total of 20 models: three standard transformers (Qwen2.5 [48], Ministral3 [31], and Llama3 [13]) and two hybrids (Qwen3.5 [45] and Gemma3[44]), enabling a controlled comparison of sparsity behavior across architectural variants.

Together, these experiments probe sparsity across six key axes, yielding the following observations:

- **Scale:** Performance under top-k sparsity improves with model size; the gap between sparse and dense decoding largely closes at scale (Figure 1).
- **Architecture:** Hybrid models tolerate sparsity better than standard transformers. Notably, at high sparsity levels (e.g., $50\times$ reduction), hybrid models maintain strong performance. We further explore the limits of extreme sparsity in fig. 2, where performance remains stable well beyond conventional sparsity regimes. In fact for larger hybrid model Qwen3.5-27B, even 16-32 tokens in top-k are enough to achieve parity with the dense model on RULER-HARD.
- **Context Length:** The qualitative behavior of sparsity remains consistent across context lengths. Observations at 32K and 128K contexts largely align, indicating that sparsity properties generalize to long-context settings without introducing additional degradation (see fig. 1).
- **Algorithm:** We separate three independent design axes for sparse decode: (i) *granularity*: per-query, per-head selection (used throughout this paper) gives the algorithm strictly more freedom than per-query, per-layer or per-batch; (ii) *selection rule*: deterministic oracle top-k (used here as a quality ceiling), approximate top-k via a learned or hashed indexer [10, 49, 56], or stochastic estimators [9]; and (iii) the *base set* of always-retained tokens (sink and local windows). Most experiments are done with oracle-topk selection. We find that deterministic top-k degrades on smaller models. See Figure 1. In such cases, we find that stochastic indexing via vAttention closes the performance gap as shown in Figure 2. However, as model scale increases, the dependence on sparsity algorithm largely reduces with oracle-topk also performing extremely well.
- **Task complexity:** Qwen3.5-27B holds across single-hop retrieval and multi-hop QA (RULER-HARD, LOFT), mathematical reasoning (AIME 2025), and agentic coding (SWE-Bench Django). On SWE-Bench Django, the head-to-head separates two regimes (Figure 1): on the 70-task subset where all three configurations attempted, dense reaches 71.4% vs 67.1% at $5\times$ and $50\times$ sparsity; on the stricter 58-task subset where all three submitted a patch, sparse is within ~ 2 points of dense (79.3% at $5\times$, 75.9% at $50\times$, 77.6% dense). The remaining gap is harness failures (timeouts, server errors, step-limit hits), not quality (Figure 7).
- **Cost:** On SWE-Bench Django, mean turns to a resolved patch fall $67 \to 57 \to 55$ (dense, $5\times$, $50\times$) and mean tokens fall 1.34M$\to 1.14$M$\to 1.08$M; unresolved tasks drop further ($92 \to 67/70$ turns), so the agent commits or quits sooner under sparsity. Combined with the kernel-level speedups in Section 5.2, the per-task cost reduction is multiplicative. AIME-2025 generation length is also stable across sparsity (Figure 8).

5 Sparsity and Hardware

5.1 Sparsity and current hardware

A common argument against sparsity concerns its poor alignment with modern hardware, which has led to considerable debate over whether block sparsity is a necessary condition for practical

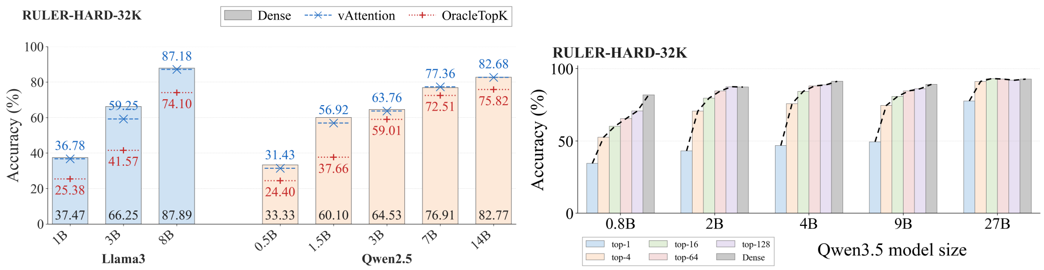

Figure 2: **Stochastic selection matches dense across model sizes; 27B hybrid tolerates 16–32 tokens. Left:** deterministic oracle top-k vs. stochastic vAttention [9] at $50\times$ sparsity on Qwen2.5, Llama3, Ministral3. vAttention is within ~ 1 point of dense at every scale; deterministic top-k degrades on smaller models, as predicted by Cor. 7 (deterministic small-budget selection cannot cover diffuse mass). **Right:** RULER-HARD-32K accuracy of Qwen3.5-{4B,9B,27B} as top-k shrinks; 32 sink and 32 local tokens kept fixed. Qwen3.5-27B matches dense at $k \in [16, 32]$.

Table 1: **Per-query, per-head irregular sparse decode is up to $76\times$ faster than FlashInfer.** Speedup over FlashInfer [52] on H100 80GB HBM3 (FP16, GQA H_q=32, H_{kv}=8, D=128, page 16, NHD, 128K context). $K\times$: each query-head attends to $1/K$ tokens; $<1\times$ is overhead. Break-even at $10\times$ for every batch; 10–$76\times$ in the 50–$500\times$ regime. No block structure imposed.

B	FlashInfer (ms)	2x	4x	10x	20x	50x	100x	200x	500x
1	0.19	0.32×	0.63×	1.45×	2.58×	5.57×	10.25×	11.05×	11.14×
4	0.72	0.33×	0.66×	1.64×	3.18×	7.45×	13.36×	24.25×	42.04×
8	1.50	0.38×	0.77×	1.90×	3.75×	8.88×	16.82×	29.64×	76.14×
16	3.08	0.45×	0.89×	2.21×	4.35×	10.54×	20.09×	37.32×	76.77×

efficiency gains. DeepSeek-V3 Attention [30] demonstrates that token-level fine-grained sparsity can yield meaningful efficiency improvements at both prefill and decode time : a result with implications not only for inference but also for training-time reductions on long-context workloads. We take this a step further in the context of decode-time sparsity. We show that it is possible to improve upon the state-of-the-art FlashInfer[2] decoding kernels using an even finer-grained sparsity pattern: per-query, per-query-head token-level sparsity. We find this effective even in challenging settings such as Grouped Query Attention (GQA) [3] where number of query heads can be much larger than key-value heads.

We benchmark sparse decode backend kernel against a full dense decode baseline on an NVIDIA H100 80GB HBM3 GPU using fp16 precision, $H_{kv} = 8$, $H_q = 32$, head dimension $D = 128$, page size 16, and NHD layout. Table 1 shows the performance of sparse backend which computes the weighted attention given sparse index and associated weights. It shows that we can leverage even this irregular sparsity. At 50–$100\times$ sparsity, our backend delivers 5.5–$20\times$ speedup over FlashInfer across batch sizes; at extreme $500\times$ sparsity, speedup reaches $76\times$ at large batch (Table 1).

To include some form of indexing mechanism, we simulate an 8-channel (16-bit precision) Double Sparsity [49] indexer and report results in Table 2. Double Sparsity achieves up to $4.17\times$ speedup in MHA and $2.81\times$ in GQA at $100\times$ sparsity. MHA crosses break-even at $2\times$ sparsity; GQA at 10–$20\times$. A lighter indexer (HashAttention [10], PQCache [56], or low-precision Double Sparsity [49]) is expected to widen these margins.

5.2 Implications for future hardware

Sparsity reduces, but does not eliminate the memory bottleneck. Across 50–$500\times$ per-query-head sparsity, our paged-KV kernel achieves 5.5–$10\times$ speedup at $50\times$ and up to $76\times$ at $500\times$ over FlashInfer at 128K context (Table 1). The bottleneck remains memory, but at lower absolute cost and with a different access pattern: irregular per-query, per-head gathers rather than contiguous loads. Current systems are poorly matched to this regime: KV caches use 16-token pages, forcing overfetch;

[2] `flashinfer.ai`

Table 2: **Sparse decode is net-positive with indexer cost included.** Speedup over FlashInfer on H100 (FP16, 128K, page 16) using Double Sparsity [49] (8×16-bit channels, untuned) for index selection, under MHA (H_q=H_{kv}=32) and GQA (H_q=32, H_{kv}=8). MHA: break-even at $2\times$, $4.17\times$ at $100\times$. GQA: break-even at $10\times$, $2.81\times$ at $100\times$. A lighter indexer (HashAttention, PQCache, low-precision Double Sparsity) is expected to widen the margin.

B	FlashInfer (ms)	2x	5x	10x	20x	50x	100x
GQA	*(H_q=32, H_k=8)*						
1	0.19	0.28×	0.56×	0.83×	1.12×	1.46×	1.65×
4	0.72	0.32×	0.67×	1.06×	1.52×	2.11×	2.45×
8	1.50	0.36×	0.75×	1.18×	1.68×	2.30×	2.66×
16	3.08	0.41×	0.85×	1.31×	1.82×	2.46×	2.81×
MHA	*(H_q=H_k=32)*						
1	0.70	0.91×	1.62×	2.18×	2.68×	3.14×	3.37×
4	2.79	1.02×	1.86×	2.56×	3.17×	3.74×	4.00×
8	5.61	1.12×	2.00×	2.71×	3.32×	3.86×	4.11×
16	11.18	1.22×	2.13×	2.84×	3.43×	3.94×	4.17×

scattered access induces TLB pressure; and indexers (e.g., HashAttention, Double Sparsity, PQCache) introduce additional reads. Thus, measured gains likely underestimate the potential of hardware designed for irregular sparse access.

Co-design: open questions for hardware. What would an accelerator built around token-level sparsity need to support natively, across training, prefill, and decode? A few questions worth asking:

- **Granularity:** KV pages were sized for contiguous access. Under irregular per-token selection, what is the right page size, and is it the same across training, prefill, and decode?
- **Top-K computation as a system primitive:** Every sparse method keeps some approximate-key state (hashing, PQ, learned). Today it lives as a per-kernel buffer that gets re-read each step. Should it instead be system-level state with its own residency tier?
- **Compute:** Sparse attention is gather followed by matmul. Should hardware fuse the two, or is software-level gather plus dense matmul good enough end-to-end?
- **Tiered KV:** Not all tokens are accessed equally. Sinks and recent tokens are read on every step; most others are read rarely. Should the memory hierarchy reflect this, instead of treating the KV cache as uniform?

6 Conclusion

The AI workload landscape is rapidly shifting toward long-context understanding and long-form generation, with tasks such as repository-scale code comprehension, long-document question answering, and agentic systems becoming increasingly common. We argue that standard attention mechanisms were not designed for such extreme context lengths. In particular, attention faces a fundamental embedding bottleneck: a relatively small hidden dimension $d \ll N$ forces information from an NNN-dimensional context to collapse into a much lower-dimensional representation. Motivated by this limitation, we envision a future in which long-context LLM inference becomes entirely sparse in the context dimension. To support this vision, we demonstrate the surprising robustness of new-generation large models to extreme sparsity, even though these models were not explicitly trained for sparse context processing. We further show that multiple forms of sparsity can already be exploited effectively on current hardware, while even greater gains may be unlocked through future hardware designs that explicitly acknowledge the inherently sparse nature of context processing. Beyond empirical evidence, we also propose a pathway toward complete context sparsity, eliminating any attention layer that attends densely over all tokens, and provide theoretical arguments supporting the feasibility of such an architecture. Overall, we believe the community should treat sparsity as a central principle when designing the next generation of model architectures, inference and training systems, and the hardware platforms on which they operate.

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A Related Works

Recent work has begun to systematically study the practical trade-offs of sparse attention for LLM inference. Nawrot et al. [34] show that sparse attention can shift the Pareto frontier, with larger sparse models outperforming smaller dense models at equivalent compute. They further identify key systems constraints: during prefilling, fine-grained token-level importance estimation is often too costly, making coarser strategies such as block-level or global selection more practical, whereas decoding allows more adaptive sparsification. Finally, they observe that tolerance to sparsity increases with sequence length, suggesting that fixed-budget or fixed top-k schemes may be misaligned with long-context regimes.

At a more fundamental level, Zucchet et al. [60] study the role of sparse attention in the emergence of model behaviors during training. They show that sparse or highly concentrated attention patterns often arise naturally over the course of optimization and may correlate with known emergent structures such as induction heads or task-specific circuits. However, the work cautions against attributing emergence solely to sparsity, noting that similar phase transitions can arise from other components (e.g., MLPs) and that large-scale emergence is tightly coupled to overall training compute (model size × steps). As a result, the causal role of sparse attention in emergence, and its interaction with scale and optimization dynamics, remains unresolved.

B Proofs

B.1 Proof of Theorem 1

Proof. Consider the linear map $T : \mathbb{R}^N \to \mathbb{R}^d$ defined by $T(a) := V^\top a$. Since $V^\top \in \mathbb{R}^{d\times N}$, we have $\operatorname{rank}(T) \le d$. Let $\mathcal{N}(T) := \{z \in \mathbb{R}^N : V^\top z = 0\}$ be the null space of T. By rank-nullity,

$$\dim \mathcal{N}(T) = N - \operatorname{rank}(T) \ge N - d.$$

We want a nonzero direction in the null space that also preserves the simplex sum constraint. Let $\mathbf{1} := (1, \dots, 1)^\top \in \mathbb{R}^N$ and consider the zero-sum hyperplane $H := \{z \in \mathbb{R}^N : \mathbf{1}^\top z = 0\}$. This hyperplane has dimension $N - 1$. By the standard dimension formula for subspaces,

$$\dim(\mathcal{N}(T) \cap H) \ge \dim \mathcal{N}(T) + \dim H - N.$$

Since $\dim \mathcal{N}(T) \ge N - d$ and $\dim H = N - 1$, we get

$$\dim(\mathcal{N}(T) \cap H) \ge (N - d) + (N - 1) - N = N - d - 1 > 0,$$

where the last inequality uses $d < N - 1$. Hence there exists a nonzero $z \in \mathcal{N}(T) \cap H$ such that $V^\top z = 0$, and $\mathbf{1}^\top z = 0$. Since $z \ne 0$, we may rescale it as $z \leftarrow \frac{z}{\|z\|_\infty}$, so that $\|z\|_\infty = 1$. This rescaling preserves $V^\top z = 0$ and $\mathbf{1}^\top z = 0$. Now, let $a_0 := \frac{1}{N}\mathbf{1}$ be the uniform attention distribution, and fix any $\beta \in (0, 1)$ and define

$$a := a_0 + \frac{\beta}{N} z, \qquad a' := a_0 - \frac{\beta}{N} z.$$

Since $\mathbf{1}^\top z = 0$, both a and a' sum to one. Moreover, since $\|z\|_\infty = 1$, each coordinate satisfies

$$\frac{1-\beta}{N} \le a_i, a'_i \le \frac{1+\beta}{N}.$$

Because $\beta \in (0, 1)$, all coordinates are strictly positive. Thus a and a' are valid attention distributions, and both have full support. They are distinct because $z \ne 0$.

Finally,

$$V^\top a = V^\top a_0 + \frac{\beta}{N} V^\top z = V^\top a_0 = V^\top a_0 - \frac{\beta}{N} V^\top z = V^\top a'.$$

Thus there exist two distinct full-support attention distributions a and a' such that

$$a \ne a', \qquad V^\top a = V^\top a'.$$

Therefore, the map $a \mapsto V^\top a$ is not injective on the attention simplex when $d < N - 1$. $\square$

B.2 Proof of Theorem 5

Proof. Write the attention outputs in matrix form as

$$o = V^\top a, \qquad \tilde{o} = V^\top \tilde{a},$$

where $a, \tilde{a} \in \mathbb{R}^N$ are the dense and sparse attention distributions, respectively.

Then

$$o - \tilde{o} = V^\top (a - \tilde{a}).$$

Taking norms and using the definition of the spectral norm,

$$\|o - \tilde{o}\|_2 \leq \|V^\top\|_2 \, \|a - \tilde{a}\|_2 = \|V\|_2 \, \|a - \tilde{a}\|_2.$$

Let $\delta := \delta(S) = \sum_{i \notin S} a_i$. Then

$$\tilde{a}_i = \frac{a_i}{1-\delta} \mathbf{1}\{i \in S\}.$$

Hence

$$(a - \tilde{a})_i = \begin{cases} -\dfrac{a_i \delta}{1-\delta}, & i \in S, \\ a_i, & i \notin S. \end{cases}$$

Therefore,

$$\|a - \tilde{a}\|_2^2 = \sum_{i \in S} \frac{a_i^2 \delta^2}{(1-\delta)^2} + \sum_{i \notin S} a_i^2.$$

We bound the two terms separately:

$$\sum_{i \in S} a_i^2 \leq (1-\delta)^2, \qquad \sum_{i \notin S} a_i^2 \leq \delta^2.$$

Substituting,

$$\|a - \tilde{a}\|_2^2 \leq \delta^2 + \delta^2 = 2\delta^2.$$

Thus

$$\|a - \tilde{a}\|_2 \leq \sqrt{2}\, \delta.$$

Combining the bounds gives

$$\|o - \tilde{o}\|_2 \leq \sqrt{2}\, \|V\|_2 \, \delta(S).$$

For top-k selection, $\delta(S) = 1 - M_k$, which yields

$$\|o - \tilde{o}\|_2 \leq \sqrt{2}\, \|V\|_2 \, (1 - M_k).$$

□

B.3 Proof of Theorem 8

Proof. Using these choices of x and y_i, Lemma 2 of [7] gives

$$\mathbb{E}\left[\left(\widehat{K}_i - K_i\right)^2\right] = \frac{1}{m} \exp\left(\frac{\|q + k_i\|_2^2}{\sqrt{d}}\right) K_i^2 \left[1 - \exp\left(-\frac{\|q + k_i\|_2^2}{\sqrt{d}}\right)\right]$$

$$\Rightarrow \mathbb{E}\left[\left(\frac{\widehat{K}_i - K_i}{K_i}\right)^2\right] = \frac{1}{m} \exp\left(\frac{\|q + k_i\|_2^2}{\sqrt{d}}\right) \left[1 - \exp\left(-\frac{\|q + k_i\|_2^2}{\sqrt{d}}\right)\right]. \tag{1}$$

Now, by Assumptions $\|q\|_2 = \|k_i\|_2 = 1$ and $|q^\top k_i| \leq \tau$, we further have $\|q + k_i\|_2^2 = \|q\|_2^2 + \|k_i\|_2^2 + 2q^\top k_i \leq 2 + 2\tau$. Therefore,

$$\mathbb{E}\left[\left(\frac{\widehat{K}_i - K_i}{K_i}\right)^2\right] \leq \frac{1}{m} \exp\left(\frac{2 + 2\tau}{\sqrt{d}}\right) \left[1 - \exp\left(-\frac{2 + 2\tau}{\sqrt{d}}\right)\right] = \frac{\exp\left(\frac{2+2\tau}{\sqrt{d}}\right) - 1}{m}. \tag{2}$$

Now using Jensen's inequality, we further have

$$
\begin{aligned}
\mathbb{E}\left|\widehat{K}_i - K_i\right| &\le \sqrt{\mathbb{E}\left[\left(\widehat{K}_i - K_i\right)^2\right]} \le K_i \sqrt{\frac{\exp\left(\frac{2+2\tau}{\sqrt{d}}\right) - 1}{m}} \\
\Rightarrow \mathbb{E}\left(\sum_{i=1}^{N}\left|\widehat{K}_i - K_i\right|\right) &\le \sqrt{\frac{\exp\left(\frac{2+2\tau}{\sqrt{d}}\right) - 1}{m}} \sum_{i=1}^{N} K_i \\
\Rightarrow \mathbb{E}\left(\frac{\sum_{i=1}^{N}\left|\widehat{K}_i - K_i\right|}{\sum_{i=1}^{N} K_i}\right) &\le \sqrt{\frac{\exp\left(\frac{2+2\tau}{\sqrt{d}}\right) - 1}{m}}. \qquad (3)
\end{aligned}
$$

Let $Z := \sum_{i=1}^{N} K_i$ and $\widehat{Z} := \sum_{i=1}^{N} \widehat{K}_i$. Therefore, $a_i = \frac{K_i}{Z}$ and $\widehat{a}_i = \frac{\widehat{K}_i}{\widehat{Z}}$, and we want to bound $\|a - \widehat{a}\|_1$. We rewrite it as

$$
\|a - \widehat{a}\|_1 = \sum_{i=1}^{N}\left|\frac{K_i}{Z} - \frac{\widehat{K}_i}{\widehat{Z}}\right| \le \sum_{i=1}^{N}\left|\frac{K_i}{Z} - \frac{\widehat{K}_i}{Z}\right| + \sum_{i=1}^{N}\left|\frac{\widehat{K}_i}{Z} - \frac{\widehat{K}_i}{\widehat{Z}}\right|. \qquad (4)
$$

The last step in eq. (4) is due to triangle inequality. The first term simply boils down to $\frac{\sum_{i=1}^{N}\left|\widehat{K}_i - K_i\right|}{\sum_{i=1}^{N} K_i}$. For the second term

$$
\begin{aligned}
\sum_i \left|\frac{\widehat{K}_i}{Z} - \frac{\widehat{K}_i}{\widehat{Z}}\right| &= \sum_i \widehat{K}_i \left|\frac{1}{Z} - \frac{1}{\widehat{Z}}\right| = \left|\frac{1}{Z} - \frac{1}{\widehat{Z}}\right| \sum_i \widehat{K}_i \\
= \left|\frac{1}{Z} - \frac{1}{\widehat{Z}}\right| \widehat{Z} &= \frac{|Z - \widehat{Z}|}{Z} = \frac{\left|\sum_{i=1}^{N}\left(\widehat{K}_i - K_i\right)\right|}{\sum_{i=1}^{N} K_i} \le \frac{\sum_{i=1}^{N}\left|\widehat{K}_i - K_i\right|}{\sum_{i=1}^{N} K_i} \qquad (5)
\end{aligned}
$$

Combining eq. (4) with the two two upper bounds, we have the following

$$
\|a - \widehat{a}\|_1 \le 2\,\frac{\sum_{i=1}^{N}\left|\widehat{K}_i - K_i\right|}{\sum_{i=1}^{N} K_i}. \qquad (6)
$$

Next, we derive a high-probability bound for the normalized perturbation term. Since

$$
\frac{\sum_{i=1}^{N}|\widehat{K}_i - K_i|}{\sum_{i=1}^{N} K_i} = \sum_{i=1}^{N} \frac{K_i}{\sum_{j=1}^{N} K_j}\left|\frac{\widehat{K}_i - K_i}{K_i}\right| = \sum_{i=1}^{N} a_i \left|\frac{\widehat{K}_i - K_i}{K_i}\right|,
$$

Jensen's inequality gives $\left(\frac{\sum_{i=1}^{N}|\widehat{K}_i - K_i|}{\sum_{i=1}^{N} K_i}\right)^2 \le \sum_{i=1}^{N} a_i \left(\frac{\widehat{K}_i - K_i}{K_i}\right)^2$. Taking expectations and using the relative MSE bound in eq. (2), we obtain

$$
\mathbb{E}\left[\left(\frac{\sum_{i=1}^{N}|\widehat{K}_i - K_i|}{\sum_{i=1}^{N} K_i}\right)^2\right] \le \sum_{i=1}^{N} a_i\, \mathbb{E}\left[\left(\frac{\widehat{K}_i - K_i}{K_i}\right)^2\right] \le \frac{\exp\left(\frac{2+2\tau}{\sqrt{d}}\right) - 1}{m} \sum_{i=1}^{N} a_i = \frac{\exp\left(\frac{2+2\tau}{\sqrt{d}}\right) - 1}{m}. \qquad (7)
$$

Therefore, by Markov's inequality applied to the square of the normalized perturbation, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$
\frac{\sum_{i=1}^{N}|\widehat{K}_i - K_i|}{\sum_{i=1}^{N} K_i} \le \sqrt{\frac{\exp\left(\frac{2+2\tau}{\sqrt{d}}\right) - 1}{m\delta}}. \qquad (8)
$$

Consequently, with probability at least $1 - \delta$,

$$
\|o - \widehat{o}\|_2 = \|V^\top (a - \widehat{a})\|_2 \le \|V\|_2 \|a - \widehat{a}\|_1 \le 2\sqrt{\frac{\exp\left(\frac{2+2\tau}{\sqrt{d}}\right) - 1}{m\delta}}\, \|V\|_2. \qquad (9)
$$

$\square$

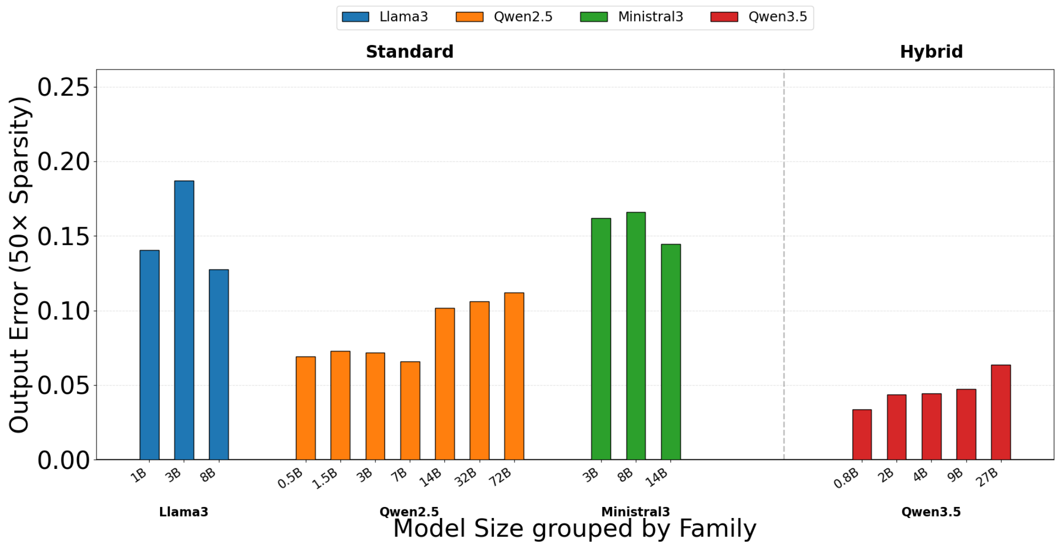

Figure 3: **Per-layer top-k error vs. sparsity, LOFT-32K.** Relative ℓ_2 error $\|o - \widetilde{o}\|_2 / \|o\|_2$ between dense SDPA and oracle top-k output. Tracks the missed-mass bound of Thm. 5. Hybrid and larger models have lower error at every sparsity level.

Figure 4: **Per-layer top-k error vs. sparsity, LOFT-128K.** Same protocol as Fig. 3. Curves overlap the 32K case; the missed-mass bound (Thm. 5) is context-length-independent in this regime.

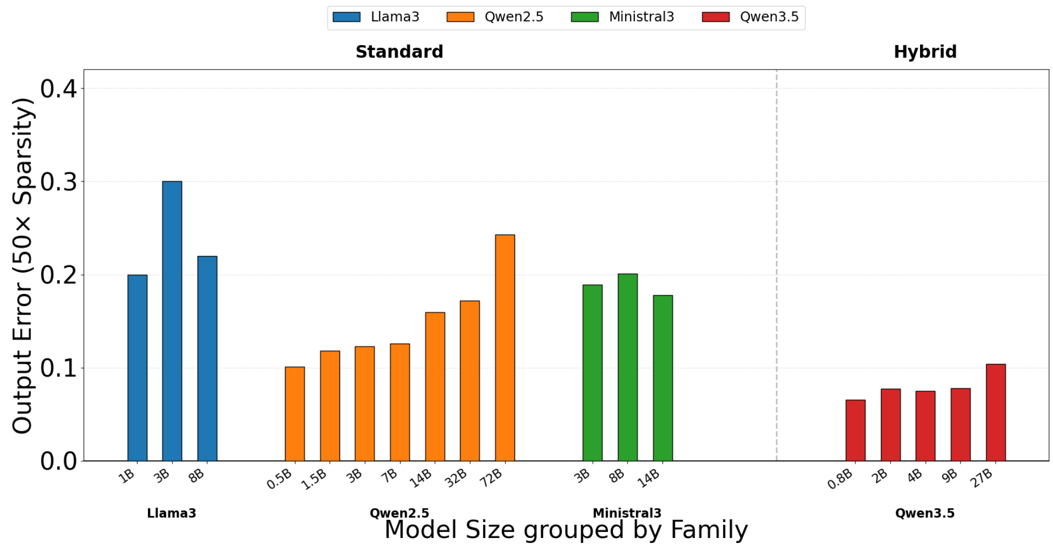

Figure 5: **Per-layer top-k error vs. sparsity, RULER-HARD-32K.** Same protocol as Figs. 3–4. Task-level quality in Fig. 1 reflects small per-layer attention error, not task-specific compensation.

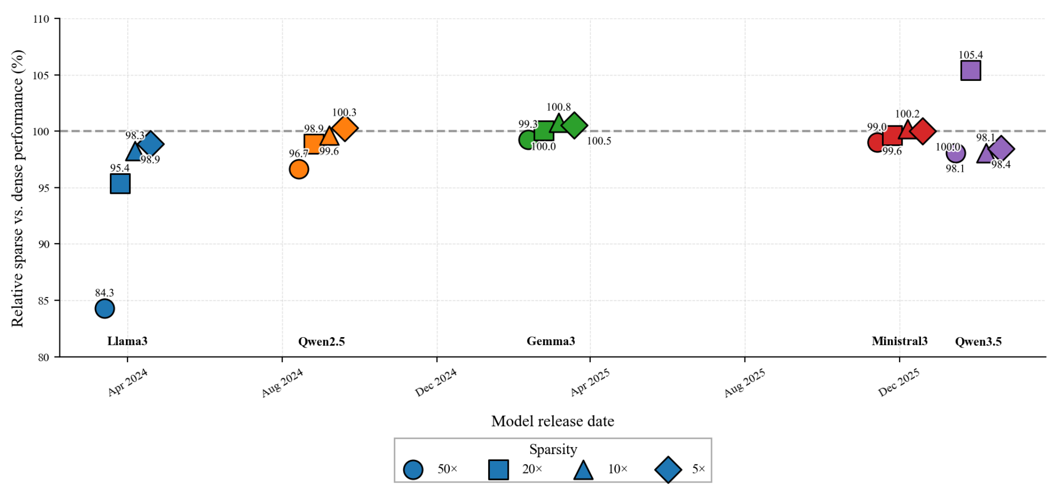

Figure 6: **Sparsity tolerance improves monotonically with release date.** One marker per (model, sparsity) point: x = release date, y = sparse-vs-dense relative score (%) on RULER-HARD-32K. Color = family; shape = sparsity ratio. Llama3 (Apr 2024): 84% at $50\times$. Every model from Aug 2024 onwards: $\geq 95\%$ at $50\times$. Qwen3.5: 105% at $20\times$.

Table 3: **Sparse decode vs. FlashInfer, 32K context.** Same kernel and config as Tab. 1; context length = 32K. Same trend; smaller absolute gains at high sparsity since the 32K dense baseline is cheaper.

Batch	2×	5×	10×	20×	50×	100×	200×	500×
1	0.40×	0.73×	1.50×	2.13×	2.59×	3.09×	3.50×	3.50×
4	0.36×	0.70×	1.61×	3.01×	6.61×	10.82×	10.95×	11.08×
8	0.40×	0.78×	1.88×	3.55×	7.67×	14.47×	20.75×	21.15×
16	0.44×	0.88×	2.13×	4.12×	9.37×	16.34×	30.88×	42.22×

Table 4: **Sparse decode vs. FlashInfer, 64K context.** Same setup as Tab. 3; context length = 64K. Reaches $81.8\times$ at $B{=}16, 500\times$; interpolates between 32K and 128K (Tab. 1).

Batch	2×	5×	10×	20×	50×	100×	200×	500×
1	0.36×	0.67×	1.45×	2.49×	3.39×	3.77×	4.17×	4.40×
4	0.34×	0.68×	1.63×	3.06×	6.81×	12.77×	21.48×	21.65×
8	0.39×	0.77×	1.88×	3.68×	8.42×	15.08×	27.77×	42.32×
16	0.44×	0.89×	2.17×	4.28×	10.06×	18.75×	32.74×	81.82×

C SWE-Bench Django: subgroup, failure-mode, and cost breakdown

This appendix reports the full SWE-Bench Django evaluation behind the headline result of Sec. 4. We compare three sparsity levels on Qwen3.5-27B: dense (100% density), oracle-top-20 ($\sim$22%), and oracle-top-2 ($\sim$3.8%), with $\leq$ 100 agent turns per task. SWE-Bench outcomes are labelled `resolved` (patch applied and target tests pass), `unresolved` (patch applied but tests fail), `empty_patch` (model produced no diff, or only test-file changes which the harness strips before applying), and `error` (patch produced but the harness could not score it; most commonly `git apply` rejected a malformed hunk).

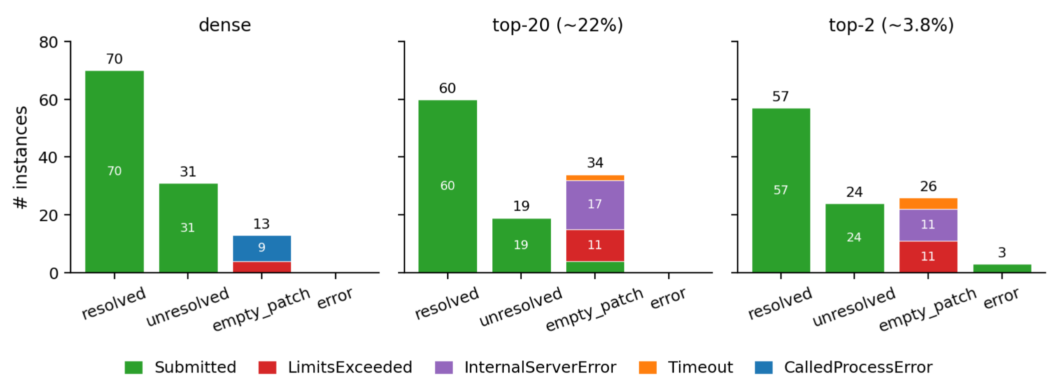

Figure 7: **Failure-mode composition shifts from model-output to infrastructure under sparsity.** Outcome counts split by exit status. Dense `empty_patch` ($n{=}13$) is dominated by `CalledProcessError` (9; `git apply` rejected the diff): the model produced a patch the harness could not apply. Sparse `empty_patch` is dominated by harness/API failures: top-20 has 17 `InternalServerError` + 11 `LimitsExceeded` + $\sim$ 2 `Timeout`, and top-2 has 11 `InternalServerError` + 11 `LimitsExceeded` + 4 `Timeout`. The shift from model-output failures (`CalledProcessError`) to harness failures (`InternalServerError` / `LimitsExceeded` / `Timeout`) is the mechanism behind the S_0-vs-S_3 gap in Fig. **??**.

D Extensions of Theorem 8

We emphasize that the unit-norm assumption in Theorem 8 is used only to simplify the geometric factor in the FAVOR+ second-moment bound. The same argument applies under the more general bounded-norm condition, e.g., $\|q\|_2 \leq R$ and $\|k_i\|_2 \leq R$ for all i. If, in addition, the query–key

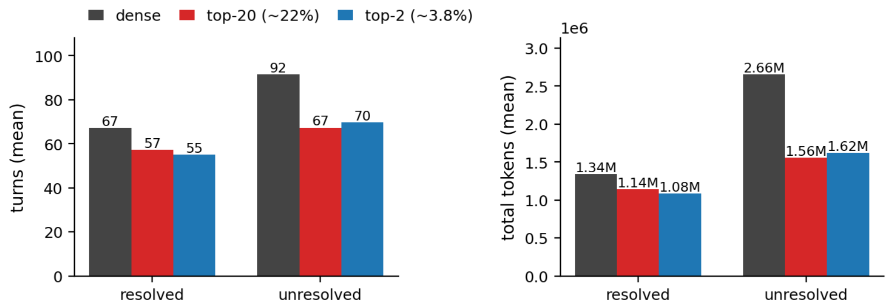

Figure 8: **Sparse decode reduces turns and tokens per task.** Mean agent turns (left) and mean total tokens (right) per outcome. For *resolved* tasks, sparse runs use $\sim 15\%$ fewer turns ($67 \to 57 \to 55$) and ~ 15–19% fewer tokens (1.34M$\to$ 1.14M$\to$ 1.08M). For *unresolved* tasks, the reduction is larger ($92 \to 67/70$ turns, 2.66M$\to$ $1.56/1.62$M tokens), suggesting the agent commits or quits sooner under sparsity. Combined with the per-decode kernel speedups in Sec. 5.2, the per-task cost reduction is multiplicative. Cost on `empty_patch` / `error` outcomes (omitted here, they are dominated by harness failures) in Tab. 6.

Table 5: **Resolution rate by subgroup (resolved/total).**

Subgroup	n	dense	top-20 ($\sim$22%)	top-2 ($\sim$3.8%)
S_0 union	114	$70/114 = 61.4\%$	$60/113 = 53.1\%$	$57/110 = 51.8\%$
S_1 3-way intersection	109	$66/109 = 60.6\%$	$57/109 = 52.3\%$	$56/109 = 51.4\%$
$S_2 = S_1 \cap$ no errors	106	$64/106 = 60.4\%$	$55/106 = 51.9\%$	$56/106 = 52.8\%$
$S_3 = S_2 \cap$ all submitted	58	$45/58 = 77.6\%$	$46/58 = 79.3\%$	$44/58 = 75.9\%$

alignments satisfy $|q^\top k_i| \le \tau$ with $0 \le \tau \le R^2$, then the factor $\exp(2+2\tau/\sqrt{d})$ is replaced by $\exp(2R^2+2\tau/\sqrt{d})$. By contrast, a norm-only FAVOR+ bound that does not use the alignment condition would give the looser factor $\exp(4R^2/\sqrt{d})$. Thus, when $\tau \ll R^2$, the alignment-aware bound is substantially tighter than the crude norm-only bound. This is the regime relevant to aggregation-like attention: the query and keys may have nontrivial norms, but no small subset of keys is strongly aligned with the query.

Connection to attention entropy. For a probability distribution $p = (p_1, \ldots, p_N)$, let $H(p) := -\sum_{i=1}^{N} p_i \log p_i$ denote its entropy. The bounded-alignment assumption in Theorem 8 directly implies

$$\max_i s_i - \min_i s_i \le \frac{2\tau}{\sqrt{d}},$$

so the softmax weights satisfy

$$\frac{e^{-2\tau/\sqrt{d}}}{N} \le a_i \le \frac{e^{2\tau/\sqrt{d}}}{N} \quad \text{for all } i,$$

and consequently

$$H(a) \ge \log N - \frac{2\tau}{\sqrt{d}}.$$

Bounded query–key alignment therefore implies the softmax distribution is close to uniform, hence diffuse in the sense of Definition 4. The entropy bound is only a diagnostic consequence; the main point is that no key is sharply separated from the rest.

Table 6: **Numerical version of Fig. 8.** Mean agent turns and mean total tokens per outcome. n columns report instance counts as (dense / top-20 / top-2).

		turns (mean)			total tokens (mean)		
outcome	n (d/20/2)	dense	top-20	top-2	dense	top-20	top-2
resolved	70 / 60 / 57	67	57	55	1.34M	1.14M	1.08M
unresolved	31 / 19 / 24	92	67	70	2.66M	1.56M	1.62M
empty_patch	13 / 34 / 26	250	138	192	14.79M	6.28M	9.44M
error	0 / 0 / 3	–	–	55	–	–	1.14M